PYTHON PROGRAMMING INTERNSHIP

Task 1: Python Programming Fundamentals, Development Environment Setup & Problem-Solving Essentials

MiniProject:
Personal Expense Tracker

Submitted By :

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1. Introduction

This internship task focuses on learning the fundamentals of Python programming and developing problem-solving skills. It includes basic Python concepts such as variables, data types, operators, conditional statements, loops, functions and modules. The task also involves setting up the Python development environment using Visual Studio Code and applying the learned concepts to develop a command-line mini-project.

2. Objectives

- To understand the basic concepts of Python programming.
- To learn how to use Python with Visual Studio Code.
- To practice variables, data types, conditions, loops, functions and modules.
- To develop problem-solving skills using Python.
- To create a menu-driven Personal Expense Tracker application.

3. Development Environment Setup

The development environment was set up using Python 3.12 and Visual Studio Code. The Python extension, Pylance and Code Runner were installed in Visual Studio Code to make Python programming and execution easier. The Python version was verified using the terminal before starting the programs.

Tools Used

- Python 3.12
- Visual Studio Code
- Python Extension
- Pylance
- Code Runner
- Git and GitHub

4. Basic Python Programs

Several basic Python programs were developed to practice fundamental programming concepts such as variables, data types, user input, operators, conditions, loops, functions and modules.

Basic Python Programs List

- Hello World program
- Variables and data types
- User input and output
- Basic calculator
- Temperature converter
- Simple interest calculator
- Conditional statements
- Loops
- Functions
- Python modules

5. Mini Project – Personal Expense Tracker

The Personal Expense Tracker is a command-line application developed using Python. It allows the user to add expenses, view recorded expenses and calculate total expenditure through a simple menu-driven interface. Input validation is also included to handle invalid expense amounts.

Features

- Add expenses
- View expenses
- Calculate total expenditure
- Input validation
- Exit the application

Working of the Application

The application displays a menu with four options: Add Expense, View Expenses, Calculate Total and Exit. The user selects an option by entering a choice. When adding an expense, the user enters the amount and category. The expense is stored in a list. The application can display all recorded expenses and calculate their total expenditure.

6. Project Structure

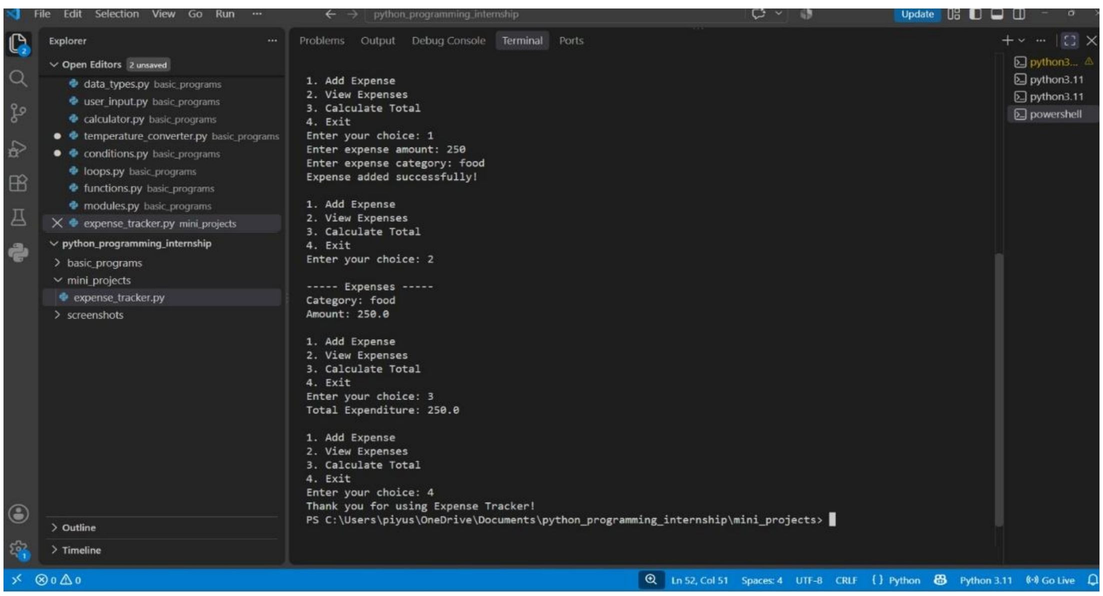
The project is organized into separate folders for basic Python programs, the mini-project and screenshots.

```
python_programming_internship/
├── basic_programs/
├── mini_projects/
│   └── expense_tracker.py
├── screenshots/
└── README.md
```

7. Testing and Output

The Expense Tracker was tested by adding an expense, viewing the recorded expense and calculating the total expenditure. Input validation was also tested by entering an invalid amount. The application successfully displayed the appropriate messages and calculated the total expenditure.

Screenshot:



8. Conclusion

This task helped in understanding the fundamentals of Python programming and applying them to a practical problem. The Personal Expense Tracker provided hands-on experience with variables, lists, dictionaries, loops, conditions, functions/modules concepts, user input and exception handling. The project also provided experience in using Visual Studio Code, Git and GitHub for development and project management.